

WOODWARD FIELD, SC, USA

WMO: 720603

Lat: 34.283N

Lon: 80.567W

Elev: 302

StdP: 14.54

Time Zone: -5.00 (NAE)

Period: 09-19

WBAN: 00195

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 22.9	(c) 27.1	(d) 4.5	(e) 7.0	(f) 30.8	(g) 9.9	(h) 9.2	(i) 34.1	(j) 20.3	(k) 52.2	(l) 18.1	(m) 50.5	(n) 1.9	(o) 290	(p) 0.374

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.6	97.4	77.0	95.0	76.1	92.7	75.5	81.4	89.3	79.4	88.4	78.3	87.6	7.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.3	153.9	84.7	77.3	143.6	82.4	75.3	134.2	81.4	45.3	89.7	43.2	88.1	41.7	89.4	92.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.2	14.8	12.3	16.0	101.0	4.7	4.1	12.6	104.0	9.8	106.4	7.2	108.7	3.8	111.7
			14.6	84.5	4.8	5.0	11.2	88.1	8.4	91.0	5.7	93.8	2.2	97.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	64.2	44.1	48.5	54.9	64.7	73.2	79.5	82.2	80.1	75.8	64.6	53.2	49.0
	DBStd	15.22	9.89	9.76	9.40	7.76	6.28	4.34	3.51	4.14	5.46	7.88	8.34	8.57
	HDD50	624	240	136	57	3	0	0	0	0	0	2	59	127
	HDD65	2531	650	466	333	98	13	0	0	0	3	104	363	501
	CDD50	5823	57	95	210	445	721	884	999	934	773	454	154	97
	CDD65	2252	1	5	21	90	269	434	534	469	325	91	8	5
	CDH74	22257	7	39	229	928	2541	4519	5700	4578	2867	773	65	11
	CDH80	9970	0	2	37	262	1045	2185	2843	2135	1234	211	16	0
(14) Wind	WSAvg	4.4	4.2	4.9	5.2	5.2	4.6	4.3	4.2	3.5	4.2	4.1	4.0	4.0
(15) Precipitation	PrecAvg	45.8	4.4	3.6	4.3	2.9	3.0	4.4	5.0	4.8	3.7	3.1	3.0	3.2
	PrecMax	61.6	9.0	7.5	10.2	6.6	7.2	10.3	9.9	10.8	8.8	16.9	8.5	8.0
	PrecMin	30.6	0.7	0.8	0.4	0.2	0.6	0.6	0.4	0.6	0.2	0.0	0.3	0.4
	PrecStd	7.3	2.0	1.8	2.0	1.7	1.7	2.6	2.8	2.4	2.2	3.1	1.8	1.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	74.7	79.0	83.9	90.1	97.2	99.6	100.2	101.8	96.8	90.3	80.7	75.1
		MCWB	63.9	64.7	65.0	69.6	74.6	75.4	78.1	83.1	74.1	74.5	69.5	70.3
	2%	DB	68.4	73.4	80.7	86.0	92.7	96.6	97.4	95.2	92.8	84.4	74.7	71.9
		MCWB	60.0	62.0	63.5	67.8	73.4	75.7	77.7	77.4	73.2	71.2	63.6	66.9
	5%	DB	64.3	70.1	75.4	82.4	89.8	93.0	95.0	92.5	90.5	81.7	71.7	68.0
		MCWB	58.2	60.1	61.3	67.1	72.3	75.0	76.9	76.6	73.3	69.4	62.8	63.9
	10%	DB	61.1	65.6	72.1	80.5	86.0	90.6	91.5	90.3	87.9	78.6	67.7	64.1
		MCWB	55.4	57.8	59.1	65.8	71.3	74.7	76.4	76.1	73.2	68.3	61.6	59.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.2	67.1	67.1	75.5	81.7	79.3	82.4	86.3	78.8	78.0	71.8	72.6
		MCDB	70.4	73.6	77.2	79.6	86.9	92.6	91.3	95.6	89.8	83.1	75.5	73.0
	2%	WB	63.6	65.2	65.7	72.6	77.4	78.2	81.0	82.0	77.0	75.7	68.3	68.8
		MCDB	66.4	71.5	75.7	80.6	87.0	90.5	90.2	89.7	86.6	80.9	71.9	70.7
	5%	WB	60.7	62.5	64.3	70.4	74.9	77.2	79.3	79.7	75.7	73.7	65.5	64.5
		MCDB	63.6	68.0	72.2	78.0	83.9	88.7	89.3	86.5	84.6	78.5	68.7	65.7
	10%	WB	55.9	59.2	62.3	68.0	73.1	76.1	78.3	78.0	74.6	70.9	62.9	60.9
		MCDB	58.8	64.2	68.6	76.5	81.8	86.8	87.9	86.0	82.6	76.3	65.8	64.3
(35) Mean Daily Temperature Range	5% DB	MDBR	20.5	21.2	22.8	23.5	21.3	20.3	19.6	18.4	19.7	21.8	22.1	19.2
		MCDBR	24.9	24.6	27.3	26.4	25.9	23.3	22.7	21.3	24.0	24.1	25.9	21.0
		MCWBR	16.8	15.4	13.2	11.7	10.0	7.3	6.5	6.8	7.7	11.4	17.1	15.4
	5% WB	MCDBR	20.2	21.4	22.8	21.2	20.6	20.6	19.5	18.7	20.1	17.0	21.4	17.1
		MCWBR	16.6	15.5	12.2	11.2	9.2	7.4	7.2	7.7	7.3	9.5	17.1	15.1
(40) Clear-Sky Solar Irradiance	taub		0.319	0.329	0.365	0.412	0.473	0.502	0.552	0.522	0.440	0.395	0.343	0.316
	taud		2.499	2.467	2.395	2.252	2.133	2.093	1.975	2.047	2.263	2.360	2.463	2.545
	Ebn at Noon		283	292	288	278	261	252	239	244	260	264	271	278
	Edh at Noon		27	31	36	43	49	51	57	53	41	34	28	24
(44) All-Sky Solar Radiation	RadAvg		871	1073	1418	1779	1945	2035	1926	1752	1506	1252	957	760
	RadStd		46	115	106	131	156	142	118	84	128	151	76	69

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (68 neighbors)	+0.97	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+324	+234

Nomenclature: See separate page